

The planets

Earth is one of four relatively small rocky planets orbiting the medium-sized star that we call the Sun. Mercury and Mars are smaller than Earth, while Venus is about the same size. The other four planets are far bigger, but even the biggest, Jupiter, is one thousand times smaller than the Sun.

Mercury

Cratered world

Diameter 3,029 miles (4,875 km)

Mass compared to Earth 0.06

Average distance from Sun 36 million miles (57.9 million km)

Number of moons 0

The smallest planet, Mercury is similar to our Moon with its cratered, barren surface and broad plains of volcanic lava. Mercury is the closest planet to the Sun but, although it has scorching hot days, its nights are freezing. This is because it has virtually no insulating atmosphere to retain heat.



Mars

Rock and ice

Diameter 4,213 miles (6,780 km)

Mass compared to Earth 0.11

Average distance from Sun 142 million miles (227.9 million km)

Number of moons 2

Only half the size of Earth, Mars does not have enough gravity to cling on to much atmosphere. Water once flowed on the planet, leaving dry riverbeds and floodplains, but the water that remains is now all frozen at the planet's north and south poles, or beneath its dry, dusty, rock-strewn surface.

▼ DISTANT WORLD

Tire tracks made by the rover vehicle Spirit scar the rocky surface of Mars, 48 million miles (78 million km) from Earth. The whole planet is tinted red by iron oxide—the same material as rust.



SCALE KEY



Earth



Planet



Venus

Searing heat

Diameter 7,521 miles (12,104 km)

Mass compared to Earth 0.82

Average distance from Sun 47 million miles (108.2 million km)

Number of moons 0

Although it looks beautiful in the night sky, Venus is shrouded with thick clouds of sulfuric acid. Its dense atmosphere of mainly carbon dioxide gas traps so much heat, the lava-covered surface of the planet reaches a blistering 867°F (464°C)—a temperature hot enough to melt lead.



Earth

Living planet

Diameter 7,926 miles (12,756 km)

Mass compared to Earth 1

Average distance from Sun 93 million miles (149.6 million km)

Number of moons 1

As far as we know, Earth is the only planet in the Solar System with any form of life. This is mainly because the other planets, even the rocky ones that are similar to Earth, do not have large amounts of liquid water—they are either too hot or too cold. Earth is the only planet with oceans of water, which cover 72 percent of its surface.